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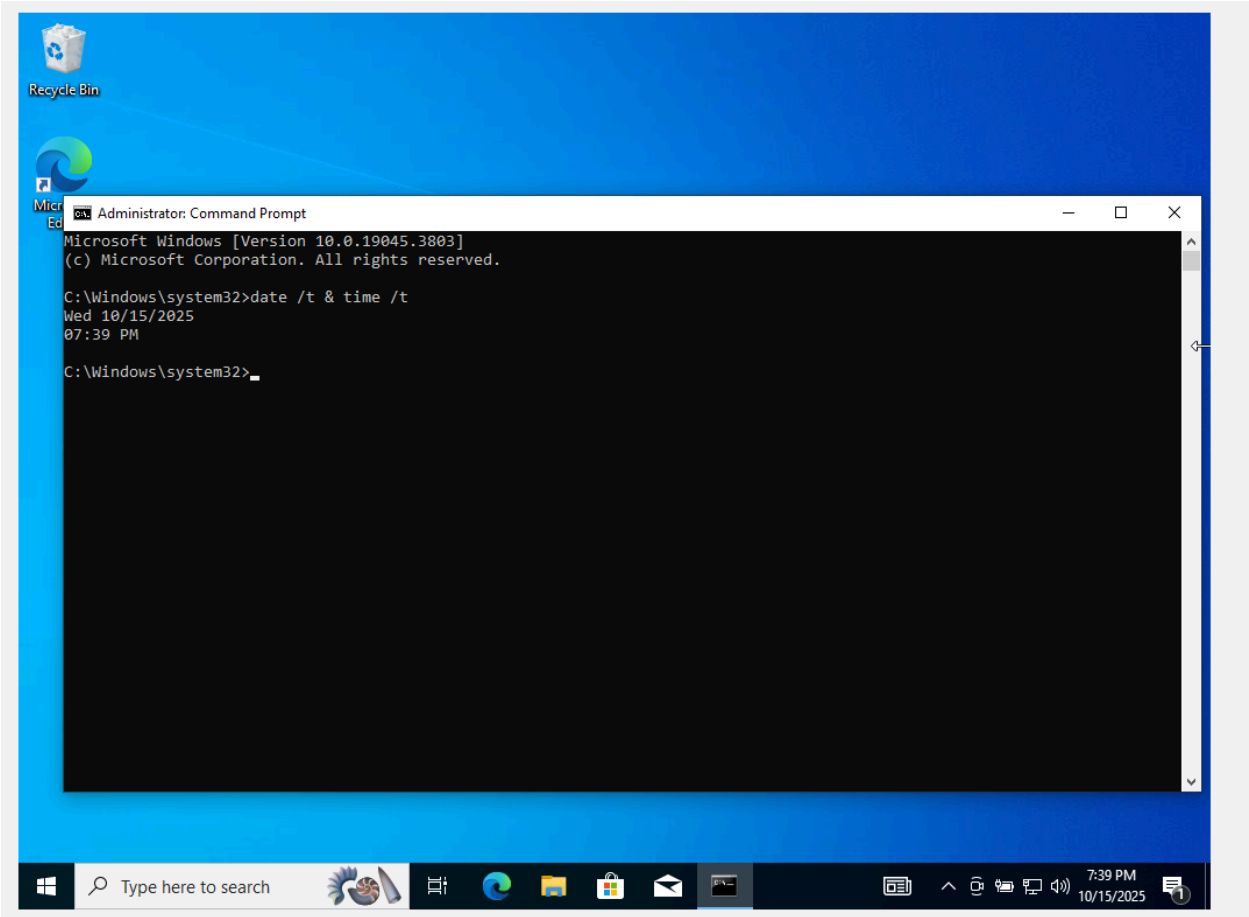
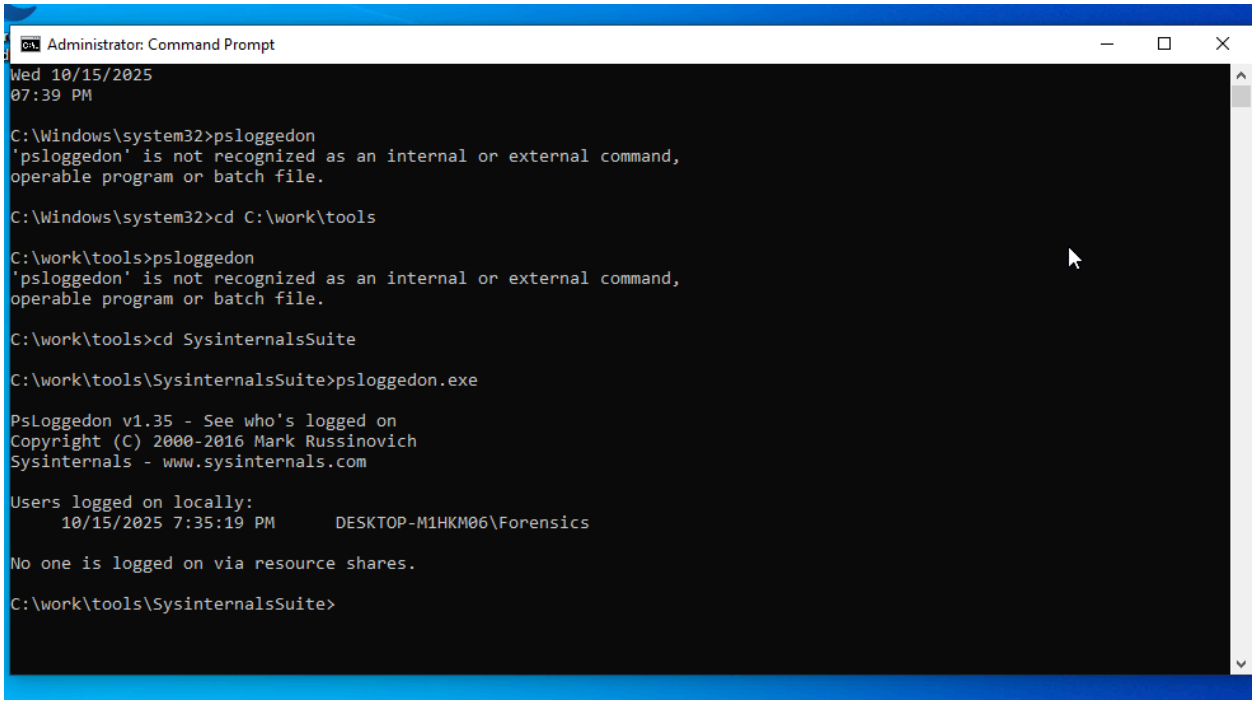
Windows Artifact Analysis

The goal of this lab was to use Windows forensic tools to collect and review live system information. I used Sysinternals utilities to identify logged-on users, sessions, active processes, and network connections. The point of this lab was to practice how to gather evidence from a running system without changing or damaging it.

Step 1: Date and Time
I started by checking the system date and time using the date /t and time /t commands. This helps timestamp all collected data. My VM showed the date and time as 10/15/2025 – 7:39 PM.

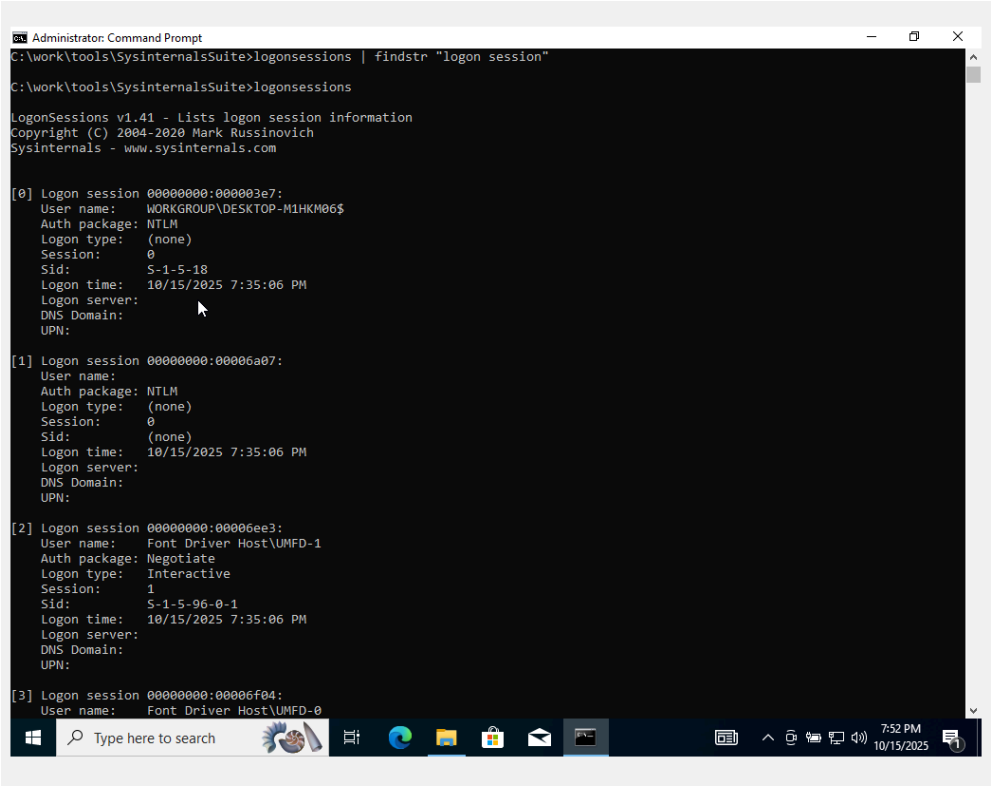


Step 2: Logged-on Users (psloggedon.exe)
Next, I opened the Sysinternals folder and ran psloggedon.exe. It showed that the currently logged-on user was Forensics on the local machine DESKTOP-M1HKM06. No one else was logged in through a resource share.

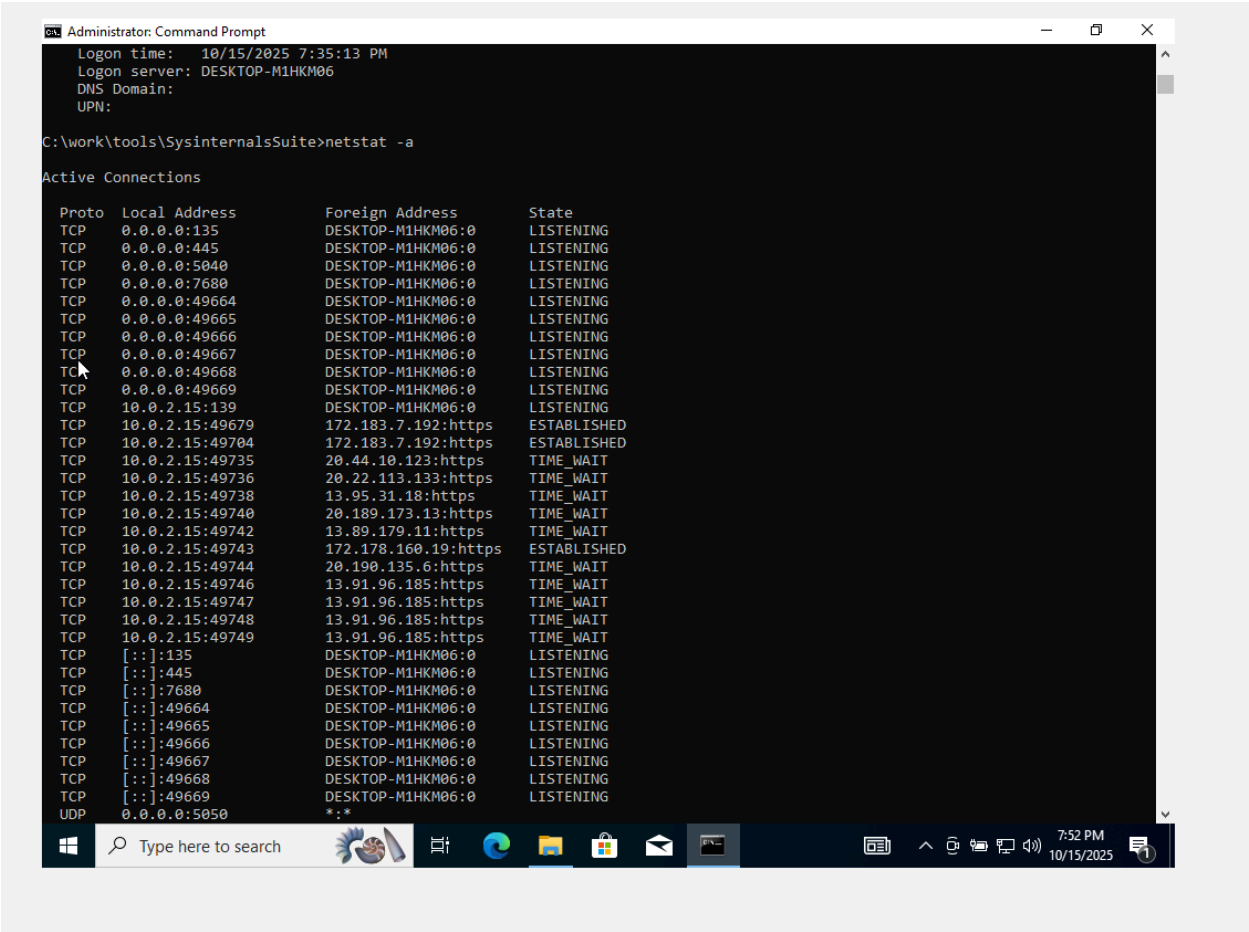


Step 3: Logon Sessions (logonsessions.exe)

I then ran logonsessions -p to see all current and previous login sessions. The output listed multiple NTLM logon sessions, including UMFD-1, UMFD-0, and an interactive session tied to the Forensics user. Each had a SID, logon time, and authentication type.



Step 4: Network Connections (netstat -a)
Using netstat -a, I viewed all open network connections and listening ports. Several ports like 445, 5040, and 7680 were in the LISTENING state, and multiple external HTTPS connections were listed as ESTABLISHED. This information helps identify active network activity or suspicious traffic.



Step 5: Processes (tasklist and pslist)
I ran the tasklist command to see all active processes, their PIDs, session names, and memory usage. Then I used tasklist /FI "PID gt 700" to filter out smaller PIDs. After that, I ran pslist -x to get detailed thread and process information. Everything appeared normal with common Windows services like svchost.exe, explorer.exe, and lsass.exe running properly.

Administrator: Command Prompt

```
RuntimeBroker.exe             5060 Console                    1      7,792 K
SgrmBroker.exe                4100 Services                   0      6,688 K
ShellExperienceHost.exe       1444 Console                    1     43,592 K
RuntimeBroker.exe             4400 Console                    1     17,016 K
cmd.exe                       1388 Console                    1      4,784 K
conhost.exe                   2916 Console                    1     24,168 K
MsMpEng.exe                   3244 Services                   0    209,808 K
MpDefenderCoreService.exe     1968 Services                   0     21,836 K
dllhost.exe                   3864 Services                   0     12,212 K
svchost.exe                   2672 Services                   0     34,076 K
svchost.exe                   4048 Services                   0     19,608 K
tasklist.exe                  2408 Console                    1      9,212 K

C:\work\tools\SysinternalsSuite>pslist -x

PsList v1.41 - Process information lister
Copyright (C) 2000-2023 Mark Russinovich
Sysinternals - www.sysinternals.com

Process and thread information for DESKTOP-M1HKM06:

Name                Pid      VM      WS    Priv Priv Pk   Faults   NonP Page
Idle                  0       8       8      60      60        9      0    0
 Tid Pri    Cswtch            State     User Time   Kernel Time   Elapsed Time
   0   0     62024          Running  0:00:00.000   0:16:41.859    0:00:00.000

Name                Pid      VM      WS    Priv Priv Pk   Faults   NonP Page
System                4    3908     136     196     216     2254      0    0
 Tid Pri    Cswtch            State     User Time   Kernel Time   Elapsed Time
  12  15        11   Wait:Executive  0:00:00.000   0:00:00.000    0:00:00.000
  16  15         1   Wait:Executive  0:00:00.000   0:00:00.000    0:00:00.000
  20  15         1   Wait:Executive  0:00:00.000   0:00:00.000    0:00:00.000
  24  16       538   Wait:Executive  0:00:00.000   0:00:00.000    0:00:00.000
  28  12        51   Wait:Executive  0:00:00.000   0:00:00.000    0:20:56.462
  32  15      7438       Wait:Queue  0:00:00.000   0:00:00.015    0:20:56.462
  36  31         1   Wait:Suspended  0:00:00.000   0:00:00.000    0:20:56.462
  40  17      1292   Wait:Executive  0:00:00.000   0:00:00.109    0:20:56.451
  44  23      1792   Wait:Executive  0:00:00.000   0:00:00.046    0:20:56.451
  48   8        12   Wait:Executive  0:00:00.000   0:00:00.000    0:20:56.451
  56  15       341    Wait:FreePage  0:00:00.000   0:00:00.000    0:20:56.422
  60   9       748    Wait:FreePage  0:00:00.000   0:00:00.328    0:20:56.421
  64   8         1    Wait:FreePage  0:00:00.000   0:00:00.000    0:20:56.421
  68   8         1    Wait:FreePage  0:00:00.000   0:00:00.000    0:20:56.421
  80  16         1       Wait:Queue  0:00:00.000   0:00:00.000    0:20:56.389
```

Type here to search 7:56 PM 10/15/2025

Administrator: Command Prompt

```
dllhost.exe                   3864 Services                   0     12,212 K
svchost.exe                   2672 Services                   0     34,048 K
svchost.exe                   4048 Services                   0     19,648 K
tasklist.exe                  4404 Console                    1      9,200 K
WmiPrvSE.exe                   256 Services                   0      9,396 K

C:\work\tools\SysinternalsSuite>tasklist /FI "PID gt 700"

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
svchost.exe                    708 Services                   0     27,524 K
fontdrvhost.exe                716 Console                    1      3,936 K
fontdrvhost.exe                724 Services                   0      2,532 K
svchost.exe                    804 Services                   0     11,704 K
dwm.exe                        896 Console                    1     47,460 K
svchost.exe                    988 Services                   0     15,772 K
svchost.exe                   1008 Services                   0     68,332 K
svchost.exe                   1016 Services                   0     22,528 K
svchost.exe                   1172 Services                   0     15,088 K
svchost.exe                   1252 Services                   0     10,288 K
Memory Compression            1284 Services                   0     86,904 K
svchost.exe                   1432 Services                   0     12,304 K
svchost.exe                   1528 Services                   0      5,876 K
svchost.exe                   1536 Services                   0      8,096 K
svchost.exe                   1572 Services                   0     15,596 K
spoolsv.exe                   1660 Services                   0     10,592 K
svchost.exe                   1692 Services                   0     15,976 K
svchost.exe                   1984 Services                   0     33,756 K
SearchIndexer.exe             2192 Services                   0     13,388 K
svchost.exe                   2212 Services                   0      7,424 K
sihost.exe                    2476 Console                    1     22,740 K
svchost.exe                   2492 Console                    1     37,396 K
MicrosoftEdgeUpdate.exe       2556 Services                   0      3,720 K
taskhostw.exe                 2600 Console                    1     13,508 K
ctfmon.exe                    2760 Console                    1     17,616 K
explorer.exe                  2008 Console                    1    112,416 K
svchost.exe                   3104 Console                    1     21,072 K
StartMenuExperienceHost.e     3576 Console                    1     50,372 K
RuntimeBroker.exe             3724 Console                    1     23,964 K
svchost.exe                   3840 Services                   0     15,856 K
SearchApp.exe                 3964 Console                    1     64,568 K
RuntimeBroker.exe             2876 Console                    1     31,524 K
svchost.exe                   4316 Services                   0      7,192 K
TextInputHost.exe             4692 Console                    1     29,100 K
```

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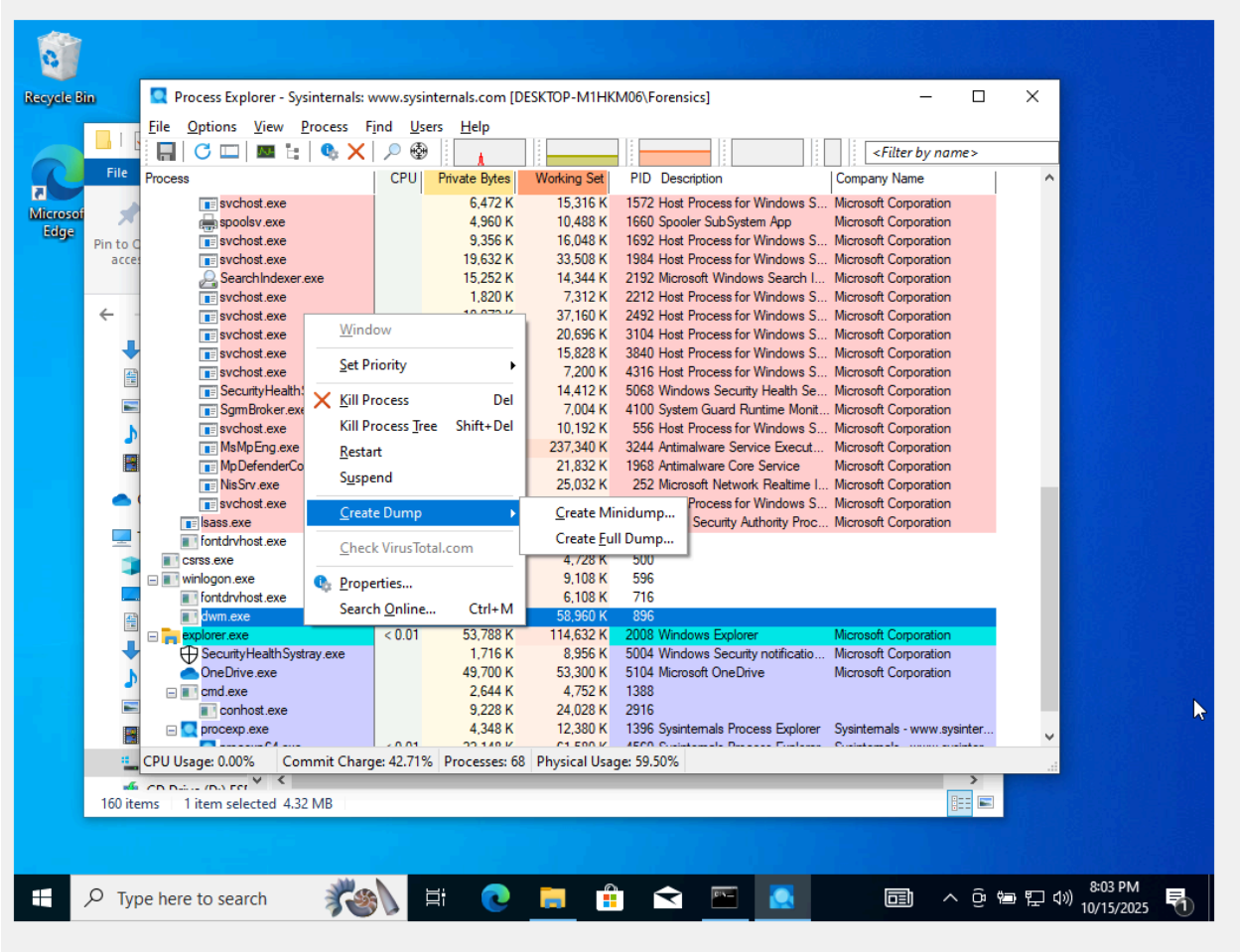
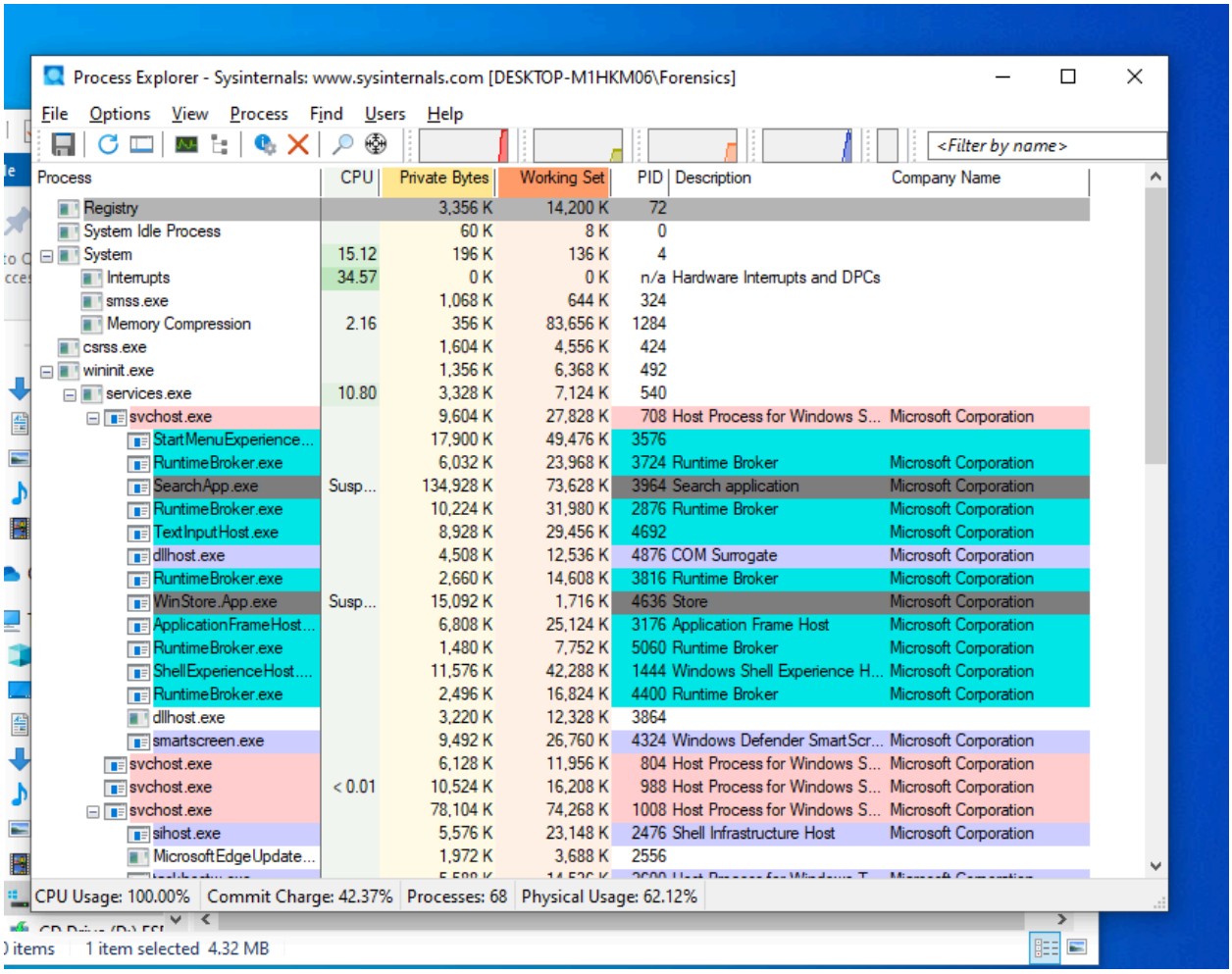
```
  UDP    10.0.2.15:137          *:*
  UDP    10.0.2.15:138          *:*
  UDP    127.0.0.1:60900        *:*
  UDP    [::]:5353              *:*
  UDP    [::]:5355              *:*

C:\work\tools\SysinternalsSuite>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0        136 K
Registry                        72 Services                   0     13,936 K
smss.exe                       324 Services                   0        644 K
csrss.exe                      424 Services                   0      4,572 K
wininit.exe                    492 Services                   0      6,376 K
csrss.exe                      500 Console                    1      4,736 K
services.exe                   540 Services                   0      7,180 K
lsass.exe                      548 Services                   0     15,612 K
winlogon.exe                   596 Console                    1      9,188 K
svchost.exe                    708 Services                   0     27,492 K
fontdrvhost.exe                716 Console                    1      3,932 K
fontdrvhost.exe                724 Services                   0      2,532 K
svchost.exe                    804 Services                   0     11,668 K
dwm.exe                        896 Console                    1     47,448 K
svchost.exe                    988 Services                   0     15,696 K
svchost.exe                   1008 Services                   0     80,476 K
svchost.exe                   1016 Services                   0     22,568 K
svchost.exe                    376 Services                   0     43,728 K
svchost.exe                    380 Services                   0     25,700 K
svchost.exe                   1172 Services                   0     15,108 K
svchost.exe                   1252 Services                   0     10,288 K
Memory Compression            1284 Services                   0     85,252 K
svchost.exe                   1432 Services                   0     12,304 K
svchost.exe                   1528 Services                   0      5,876 K
svchost.exe                   1536 Services                   0      8,136 K
svchost.exe                   1572 Services                   0     15,596 K
spoolsv.exe                   1660 Services                   0     10,592 K
svchost.exe                   1692 Services                   0     16,044 K
svchost.exe                   1984 Services                   0     33,776 K
SearchIndexer.exe             2192 Services                   0     13,296 K
svchost.exe                   2212 Services                   0      7,424 K
sihost.exe                    2476 Console                    1     22,772 K
svchost.exe                   2492 Console                    1     37,408 K
```

Step 6: Process Explorer (procexp.exe)

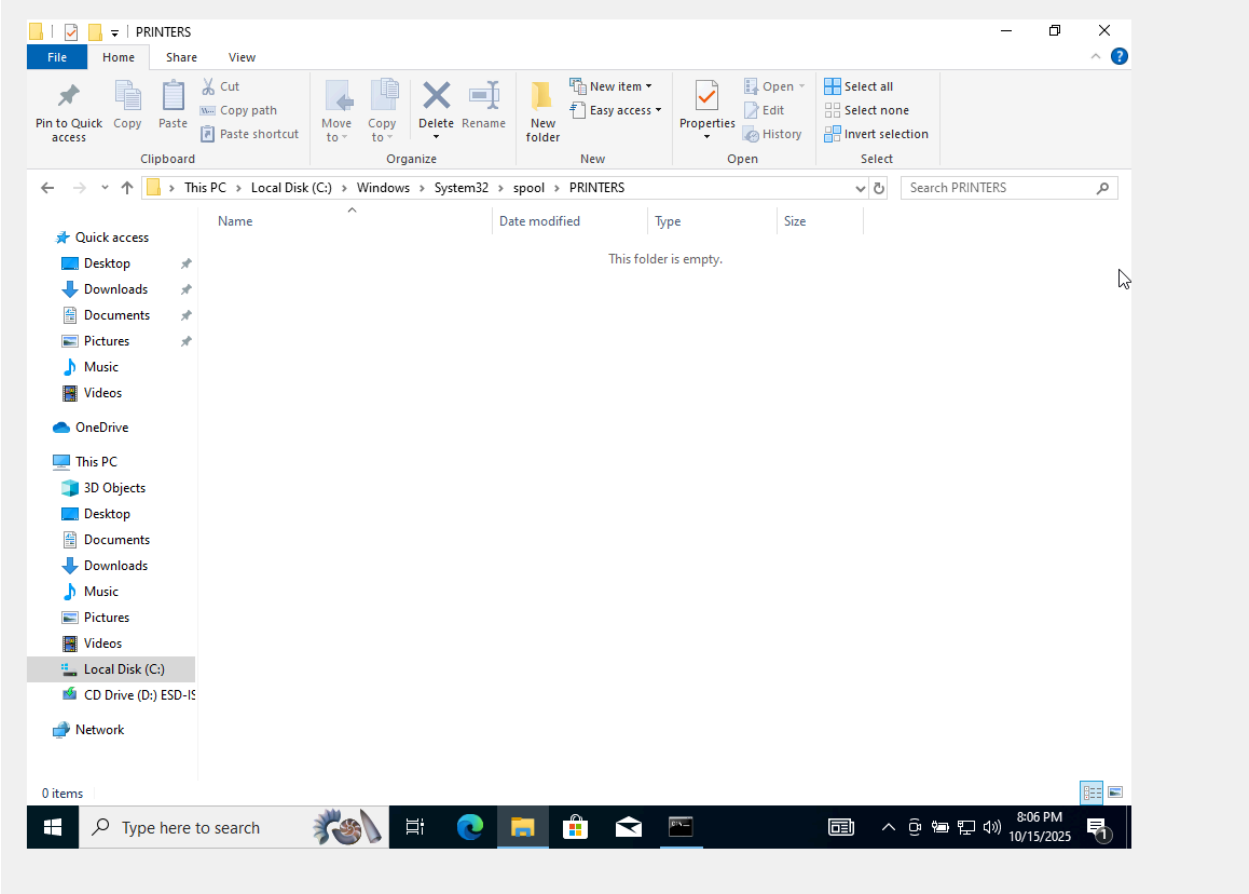
Next, I opened Process Explorer to view processes in real time. It showed CPU and memory usage, along with company names for verification. I right-clicked explorer.exe to demonstrate the Create Dump → Full Dump option, which can be used to collect process memory for deeper forensic analysis.


Recycle Bin
Microsoft Edge
Process Explorer - Sysinternals: www.sysinternals.com [DESKTOP-M1HKM06\Forensics]
File Options View Process Find Users Help
<Filter by name>
Process CPU Private Bytes Working Set PID Description Company Name
svchost.exe 6,472 K 15,316 K 1572 Host Process for Windows S... Microsoft Corporation
spoolsv.exe 4,960 K 10,488 K 1660 Spooler SubSystem App Microsoft Corporation
svchost.exe 9,356 K 16,048 K 1692 Host Process for Windows S... Microsoft Corporation
svchost.exe 19,632 K 33,508 K 1984 Host Process for Windows S... Microsoft Corporation
SearchIndexer.exe 15,252 K 14,344 K 2192 Microsoft Windows Search I... Microsoft Corporation
svchost.exe 1,820 K 7,312 K 2212 Host Process for Windows S... Microsoft Corporation
svchost.exe 37,160 K 2492 Host Process for Windows S... Microsoft Corporation
svchost.exe 20,696 K 3104 Host Process for Windows S... Microsoft Corporation
svchost.exe 15,828 K 3840 Host Process for Windows S... Microsoft Corporation
svchost.exe 7,200 K 4316 Host Process for Windows S... Microsoft Corporation
SecurityHealth 14,412 K 5068 Windows Security Health Se... Microsoft Corporation
SgrmBroker.exe 7,004 K 4100 System Guard Runtime Monit... Microsoft Corporation
svchost.exe 10,192 K 556 Host Process for Windows S... Microsoft Corporation
MsMpEng.exe 237,340 K 3244 Antimalware Service Execut... Microsoft Corporation
MpDefenderCo 21,832 K 1968 Antimalware Core Service Microsoft Corporation
NisSrv.exe 25,032 K 252 Microsoft Network Realtime I... Microsoft Corporation
svchost.exe Process for Windows S... Microsoft Corporation
lsass.exe Security Authority Proc... Microsoft Corporation
fontdrvhost.exe
csrss.exe 4,728 K 500
winlogon.exe 9,108 K 596
fontdrvhost.exe 6,108 K 716
dwm.exe 58,960 K 896
explorer.exe < 0.01 53,788 K 114,632 K 2008 Windows Explorer Microsoft Corporation
SecurityHealthSystray.exe 1,716 K 8,956 K 5004 Windows Security notificatio... Microsoft Corporation
OneDrive.exe 49,700 K 53,300 K 5104 Microsoft OneDrive Microsoft Corporation
cmd.exe 2,644 K 4,752 K 1388
conhost.exe 9,228 K 24,028 K 2916
procexp.exe 4,348 K 12,380 K 1396 Sysinternals Process Explorer Sysinternals - www.sysinter...
Window
Set Priority
Kill Process Del
Kill Process Tree Shift+Del
Restart
Suspend
Create Dump
Create Minidump...
Create Full Dump...
Check VirusTotal.com
Properties...
Search Online... Ctrl+M
CPU Usage: 0.00% Commit Charge: 42.71% Processes: 68 Physical Usage: 59.50%
160 items 1 item selected 4.32 MB
Type here to search
8:03 PM
10/15/2025



Step 7: Print Spool Folder

Finally, I opened the folder at C:\Windows\System32\spool\PRINTERS. It was empty, showing no active or saved print spool files. If files had been there, they would've shown details about recent print activity.



Conclusion

In this lab, I used different Sysinternals tools to gather live forensic data from a Windows system. I verified who was logged in, what sessions were active, which processes were running, and what network connections were open. The system appeared normal with no signs of unauthorized users or suspicious processes. This lab helped me understand how live forensic analysis can provide quick system insights without altering digital evidence.